

- 4 (a) Sound waves produced by a loudspeaker arrive at the microphone M via two different paths, LXM and LYM as shown in Fig.4.1. The left-tube is fixed in position, while the right-tube is a sliding-section.

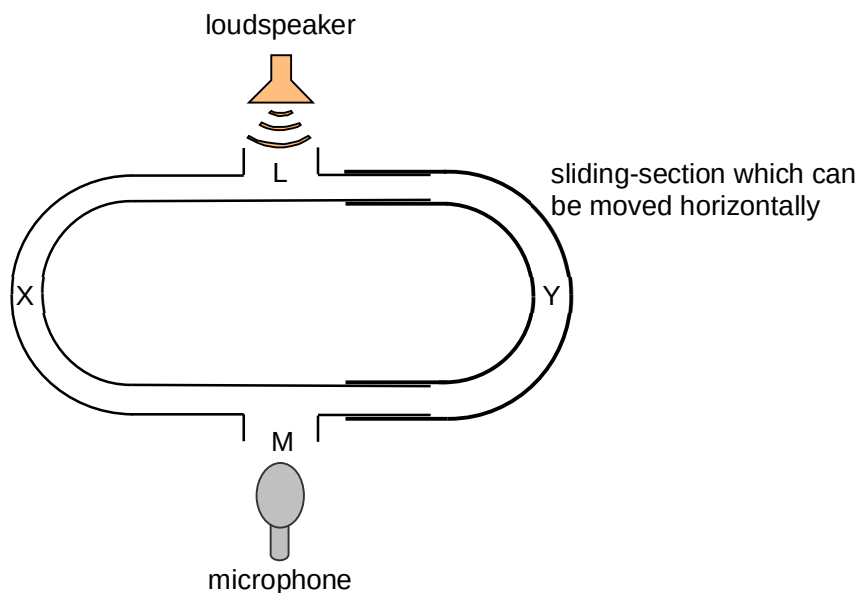


Fig. 4.1

Initially, the lengths of paths LXM and LYM are equal. The sliding-section is then pulled out horizontally to the right by a distance of 0.020 m, and the loudness detected at microphone M changes from a maximum to a minimum.

- (i) Determine the wavelength of the sound waves.

wavelength = m [2]

- (ii) When the opening at M is subsequently sealed, explain why a stationary (standing) wave can be set up in the tube.

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 [2]

- (b) Fig. 4.2 shows a monochromatic light of wavelength 600 nm passing through fine nylon threads. The threads act as a diffraction grating with lines in the horizontal direction. Part of the pattern of light showing the central bright fringe and its first two adjacent fringes on the screen is shown in Fig. 4.3.

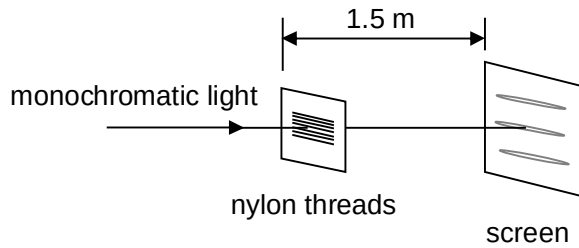


Fig. 4.2

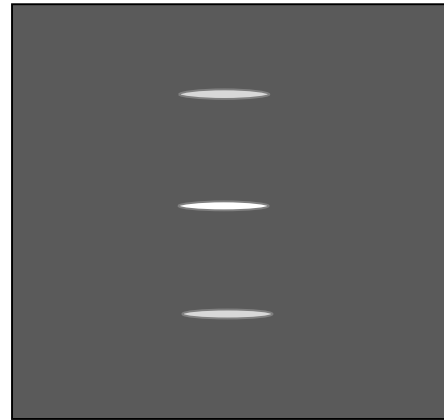


Fig. 4.3 (full size)

- (i) Calculate the angle between the two first orders of the diffracted light.

angle = ° [2]

- (ii) Using your answer to (i), determine the number of nylon threads per millimetre.

number of threads per millimetre = [2]

5 Four point charges each of magnitude q are placed at the corners of a square ABCD of